

MONTERA

— RISE BEYOND INFINITY —

CDR 2: PROOF OF CONCEPT

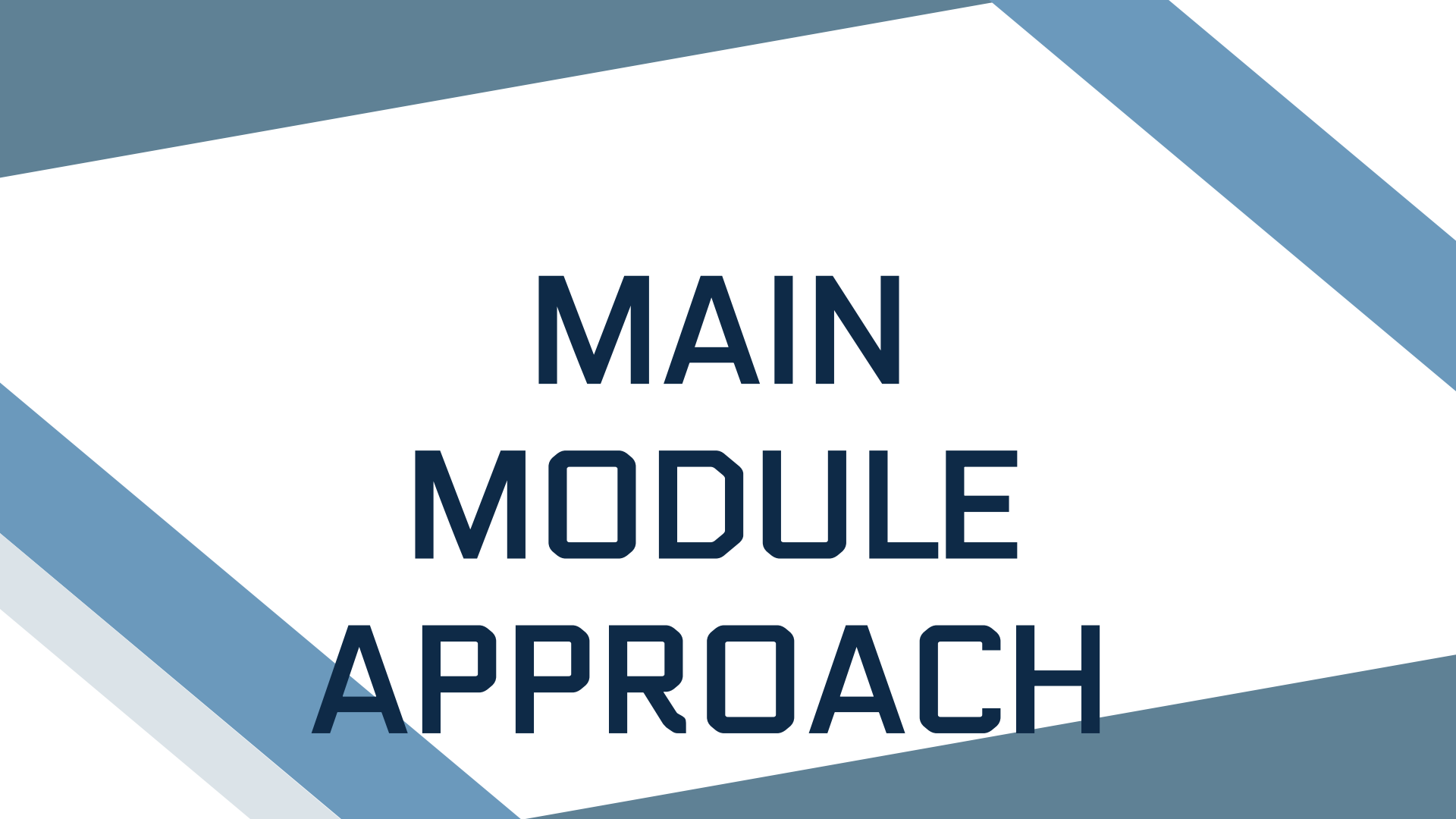
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2024 Panda-monium

The Carpa Initiative has challenged Montera to:

- **Create a self-assembling exploratory suite of robotic mechanisms (Silverdog)**
 - **Climb any surface or structure**
 - **Ascend a curved surface with a minimum diameter of 10"**
 - **Have robotic vision and vision tracking capabilities**
 - **Capable of transferring signals and electrical power between modules**
 - **Provide computer processing power**
 - **Provide a live video feed and illuminate**
 - **Be able to collect artifacts for further examination**



MAIN MODULE APPROACH

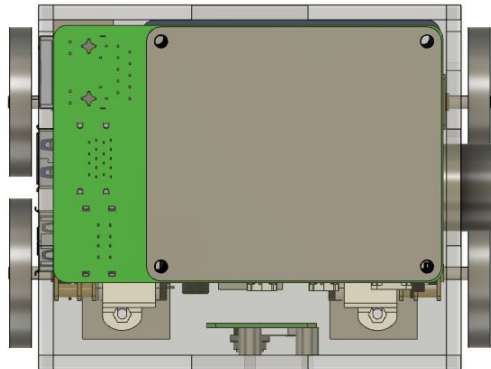


Figure 1.1: Top View

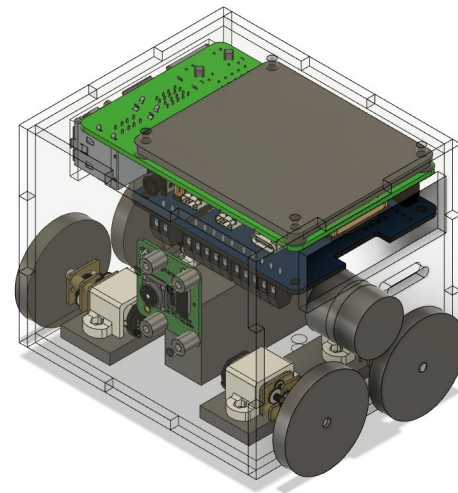


Figure 1.2: Isometric View

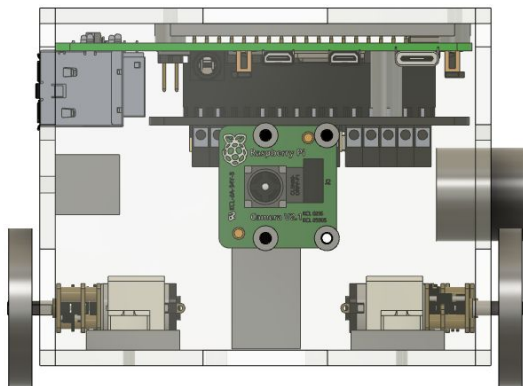


Figure 1.3: Front View

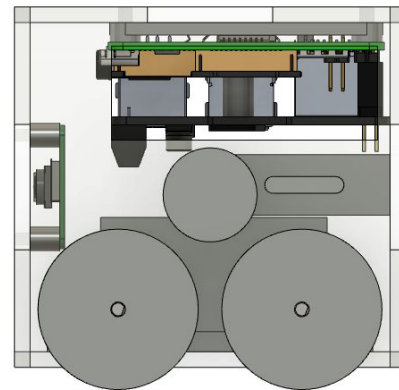
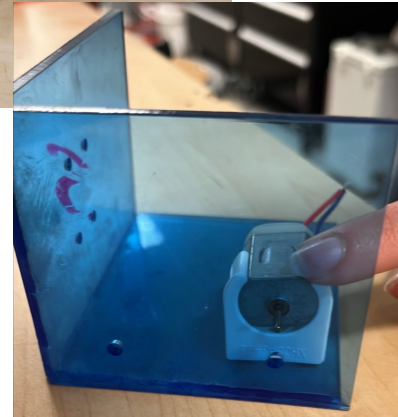
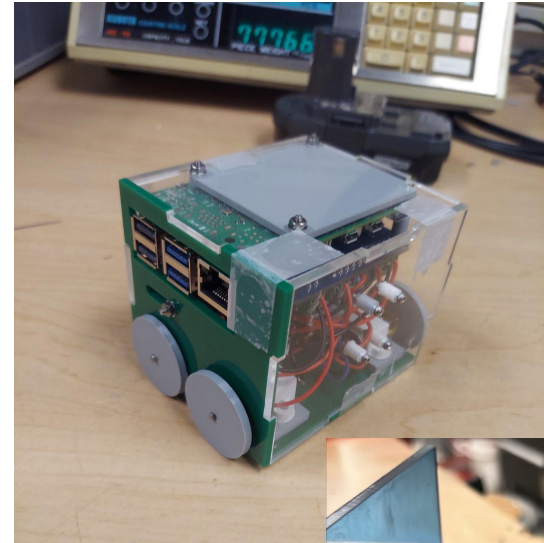


Figure 1.4: Side View

Manufacturing Explanation - Main

- Prototyping
 - Acrylic
 - Cuts in < 5 minutes
 - Lots of available 1/8" material
 - Transparent
 - Notches (Taped)
 - Tensile Strength: 75 MPa
 - Flexural Strength: 115 MPa
- Future
 - Acrylic
 - Plastic cement
 - 1/16" thickness



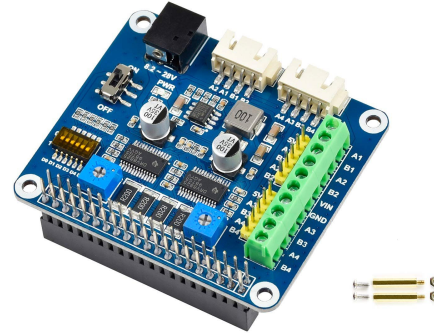
Components

Raspberry Pi Model 5B



- Controls electrical components
- Better CPU, GPU, and RAM compared to standard model

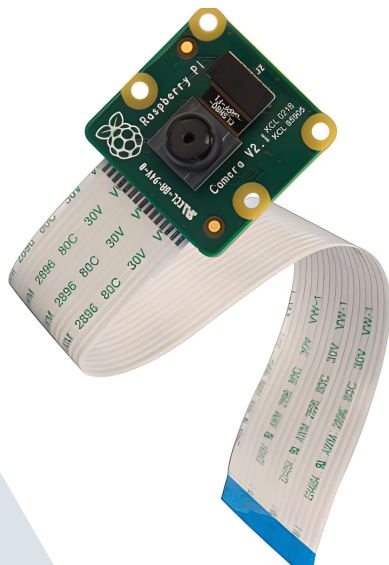
DC & Stepper Motor HAT for Raspberry Pi



- Connect to the gpio pins on the Raspberry Pi
- Control up to 4 DC motors

Components

Raspberry Pi Camera Module 3



MN21 / A23 12-Volt Battery



12V/24V to 5V DC-DC Converter



Components

**DC 12V 1000 RPM N20 High
Torque Speed Reduction
Motor**



**Motor type: DC Motor
Horsepower: 1000 RPM
Volts: 12V
Rated Current: 200mA
Weight: 9g**

**3602 Series Press-Fit
Wheel 32mm Diameter:**



- **Designed to work
with N20 High
Torque Speed
Reduction Motor**

Sister Module

- Connects to the side of the main module
- Stationary by itself
- Will transfer power + data

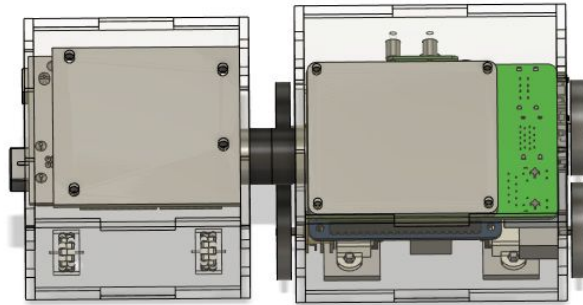


Figure 2.3: Connection of Main & Sister Modules

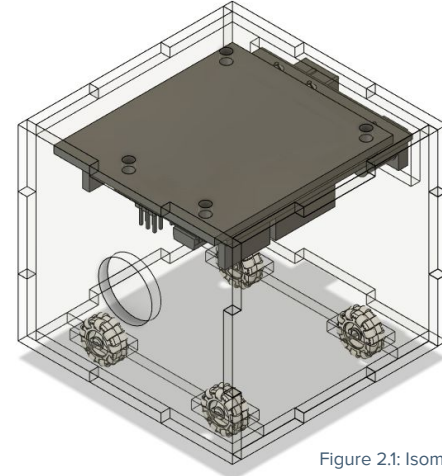


Figure 2.1: Isometric View

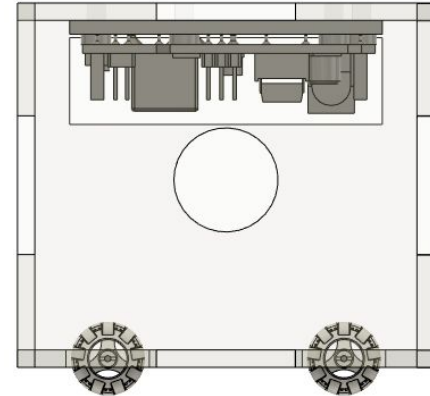
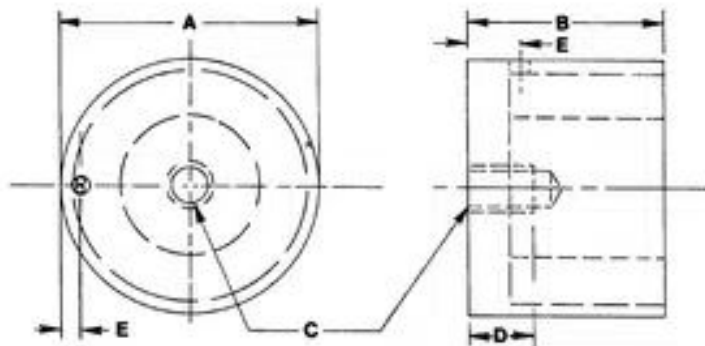


Figure 2.2: Front View

Electromagnet Connection

- Electromagnets attach main and sister modules
 - Produces strong magnetic field with a current
 - Quick modular assembly and disassembly



Part #	Diameter	Height	Thread	Thread Depth	DC Volts	Watts	Holding Force (lbs)	Wt.
BDE-080 8-12	0.75 in	0.75 in	10--32	0.25	12	0.7	6	1.50 oz

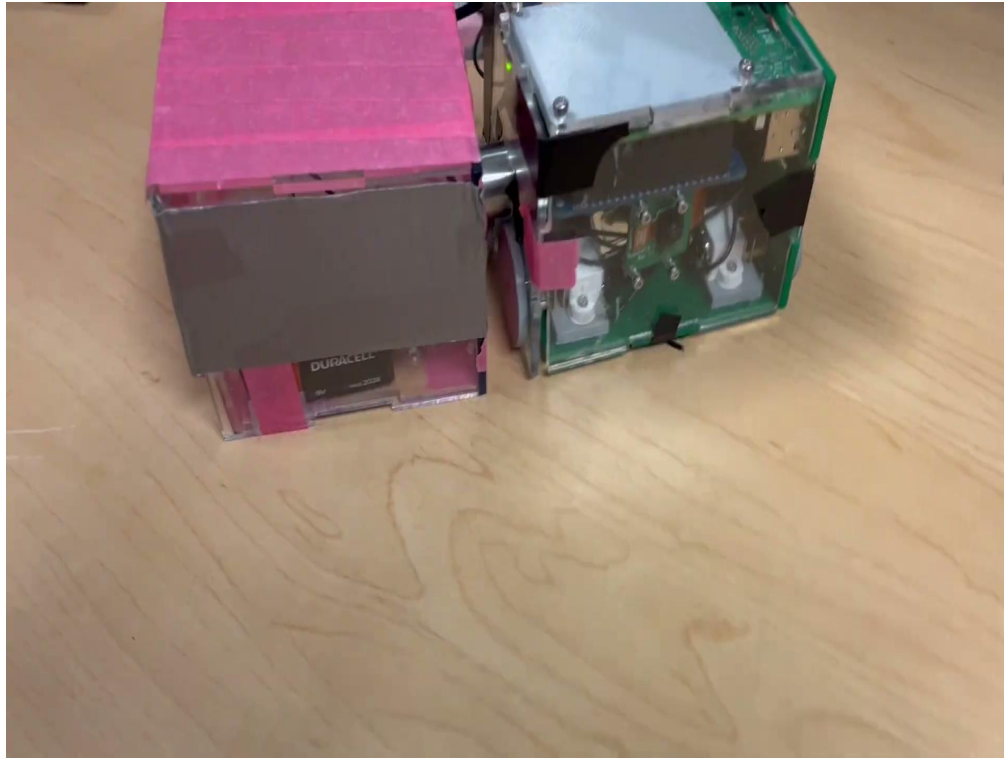
Table 1: Electromagnet Dimensions

Tests



Video 1: Main Module Test

Tests



Video 2: Main Module Test with Second Module



PROPELLER/ CLIMBING SYSTEM APPROACH

Wall Climbing System - CAD Version 1

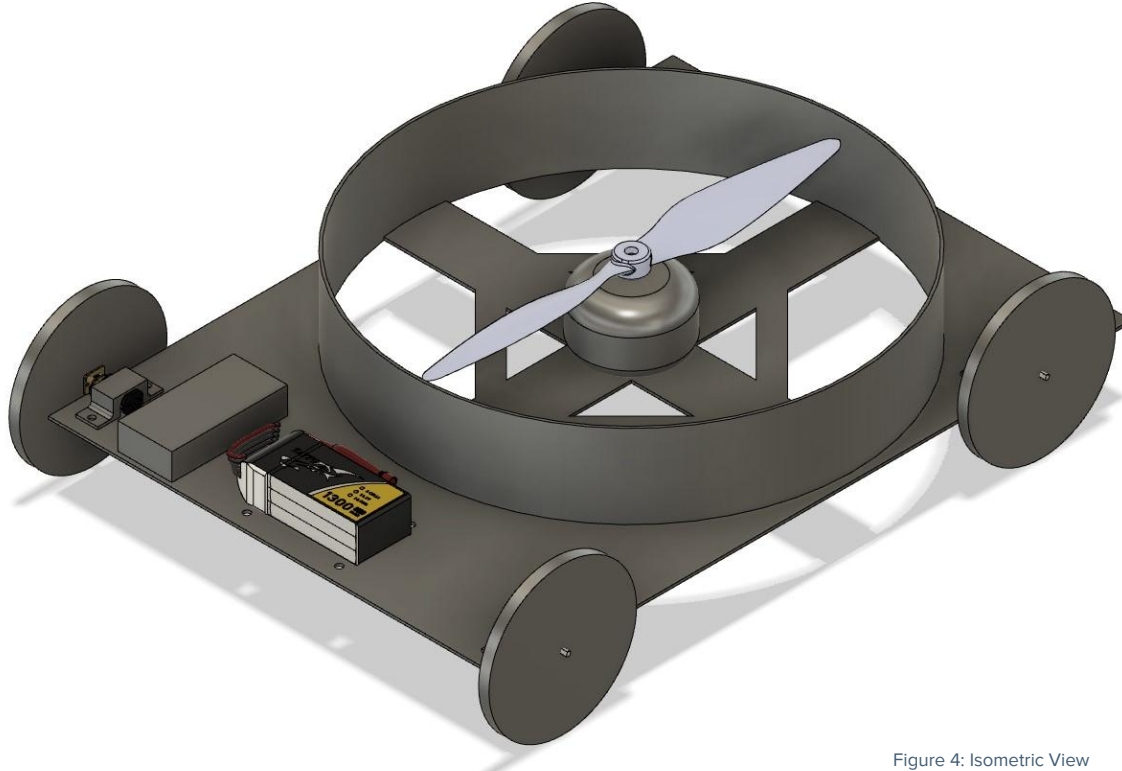


Figure 4: Isometric View

Wall Climbing System - CAD Version 2

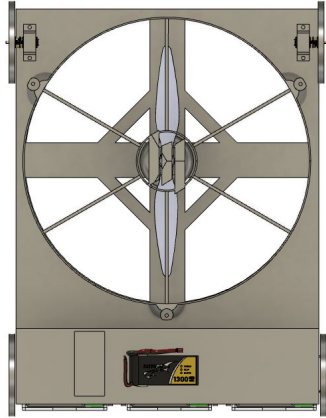


Figure 5.1: Top View

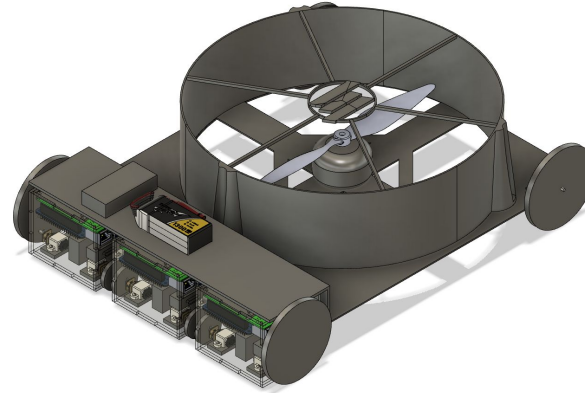


Figure 5.2: Isometric View

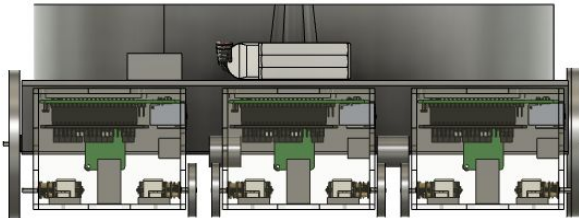


Figure 5.3: Front View



Figure 5.4: Side View

Wheels

3602 Series Press-Fit Wheel 80mm diameter:

- **Silicon treads** to provide more grip with the wall when climbing
- **Optimal because it fits well with the N20 Gear motor**

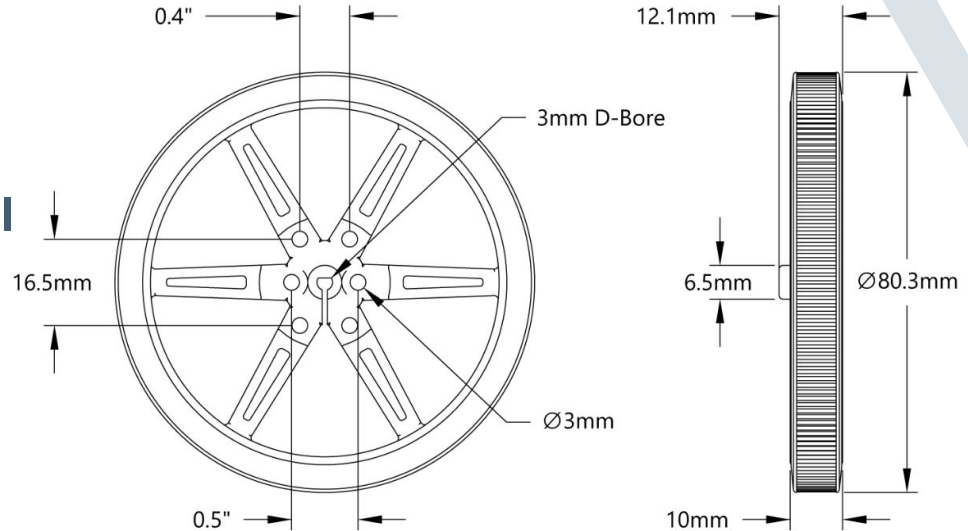


Figure 6: 3602 Series Press-Fit Wheel Dimensions

Motor Thrust Calculation

2 Scenarios:

- **Worst scenario
(slicked glass)**
- **Reasonable scenario
(concrete)**

**XOAR Titan T5015
motor may not be
optimal**

Table 2:Thrust Calculations

$$\begin{aligned}
 F_{\text{nety}} &= F_g - F_s \\
 F_{\text{nety}} &= 0 \\
 0 &= F_g - F_s \\
 F_s &= F_g = mg = 1\text{kg}(10\text{m}^2/\text{s}) \\
 10\text{N} &= F_s
 \end{aligned}$$

Thrust needed for concrete surface

Predicted Coefficient $M = 0.5$
 $(10\text{N} = 0.5F_t)/0.5$
 $20\text{N} = F_t \approx 2\text{kg thrust}$

Thrust needed for glass surface

Lowest possible $M = 0.2$
 $10\text{N} = MF_T$
 $(10\text{N} = 0.2 F_T)/0.2$
 $50\text{N} = F_T$
 $\sim 5\text{kg of Thrust needed for worst case scenario}$

Battery Calculations

- OVONIC 1300mAh 14.8V Lipo Battery (Initial)
- Battery must power:
 - 4 Hobby Motors
 - 1 Drone Motor
 - 1 Raspberry Pi
- Maximum Runtime: 1.7 Minutes
- Potential issues with runtime and battery power

Figure 7: Battery Power Calculations

<https://sciencing.com/use-variatic-5114264.html>

Power Consumption (100%):

- (4) GREARTISAN DC - $15.6W * 4 = 62.4wH$ (1hour)
 - 1.3 Amps
 - 12V
- XOAR TITAN T5015 - 1034 W
 - Data Found from Website
 - <https://www.xoarintl.com/brushless-electric-motors/titan/titan-T5000-heavy-lifting-series/#spec>

• Circuit Board – 12W

19.11 Watthours - Battery Power

.017 hours = 1.03 minute

Power Consumption (75%):

- (4) GREARTISAN DC – $11.7W * 4 = 46.8wH$ (1hour)
 - .975 Amps
 - 12V
- XOAR TITAN T5015 - 609.0 W
 - Data found from Website
 - <https://www.xoarintl.com/brushless-electric-motors/titan/titan-T5000-heavy-lifting-series/#spec>

• Circuit Board – 9W

19.11 Watthours - Battery Power

664 Total Watts Consumer

.029 hours = 1.7 minute

1st Iteration Electrical

XOAR TITAN T5015 Motor

- Motor type: Brushless
- Max. Continuous Power(W): 1034.0
- Idle Power Draw: 1.1A
- Unloaded Speed: 5,786 RPM
- Size: $\Phi 58.5 * 36.5\text{mm}$
- Thrust: 4.1 kg
- Weight: 267g



1st Iteration Electrical

OVONIC 4S Lipo Battery 80C

1300mAh 14.8V



Velineon VXL-8s
Waterproof Motor
Controller



- Input voltage: 4s/6s/8s
- LiPo Battery Compatible
- Size: 2.23 in (58 mm) x 2.83 in (72 mm) x 1.81 in (46 mm)
- Weight: 215.5 g

2nd Iteration Electrical

A2807 1500KV 6S Brushless Motor

Motor type: Brushless

Thrust: 2.2 kg

Volts: 24V(6S)

suggested ESC: 60A~70A

Max. Continuous

Power(W): 1098.8w

Weight: 56g



LiPo 2000 6S 22.2v Battery Pack



ESC 2-6S LiPo Brushless Electric Speed Controller



- Input voltage: 2s/4s/6s
- LiPo Battery Compatible
- Size: 4.65 x 3.43 x 1.06 inches
- Weight: 108.g

Manufacturing Explanation - Fan

- Prototyping
 - Wood
 - Fast production time
 - Easily broke
 - Did not cut all the way
- Future Versions
 - Acrylic
 - triangular/hexcomb design
 - Thin metal reinforcements
 - Velcro attachments

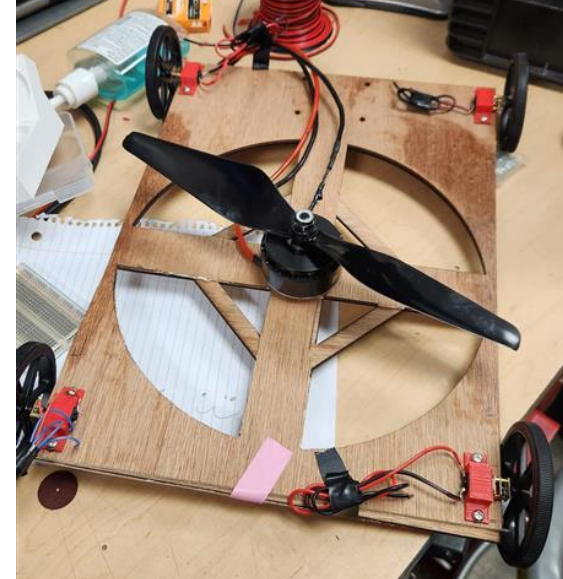
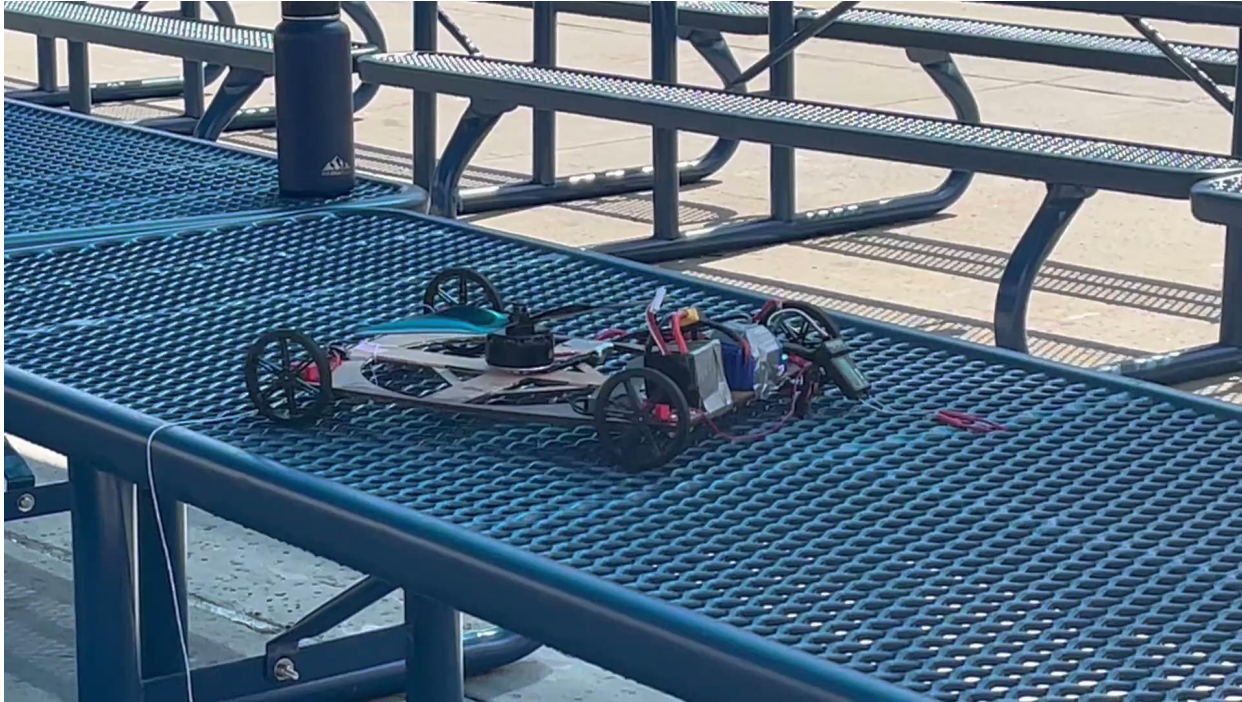


Figure 8: Climbing System Prototype

Tests



Video 3: Propeller Test

Tests

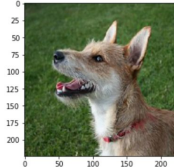


Video 4: Propeller Wall Climb Test

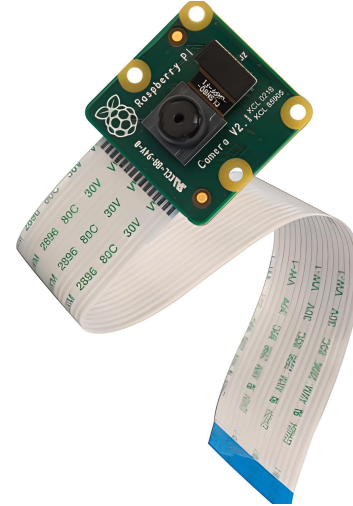
Programming Module System Sensor

- Raspberry Pi Camera feeds data to OpenCV for Image Processing
- TensorFlow machine learning uses trained data to locate modules in image
 - Controls 4 Wheel motors

```
Downloading data from https://i.ibb.co/Y2e0NH6/test-dog.jpg
16384/16291 [*****] - 0s 1us/step
1. miniature_schnauzer 23.57%
2. Scottish_deerhound 9.04%
3. silky_terrier 7.77%
4. Australian_terrier 7.40%
5. cairn 7.35%
6. standard_schnauzer 6.67%
7. Irish_wolfhound 6.01%
8. wire-haired_fox_terrier 4.93%
9. West_Highland_white_terrier 3.23%
10. Irish_terrier 3.15%
11. kelpie 2.54%
12. Ibis_hound 2.19%
13. tenniss_ball 1.60%
14. Norwich_terrier 1.59%
15. Scotch_terrier 1.49%
```



Raspberry Pi
Camera Module 3



- Example of TensorFlow being used on a dog

Programming Module System Sensor

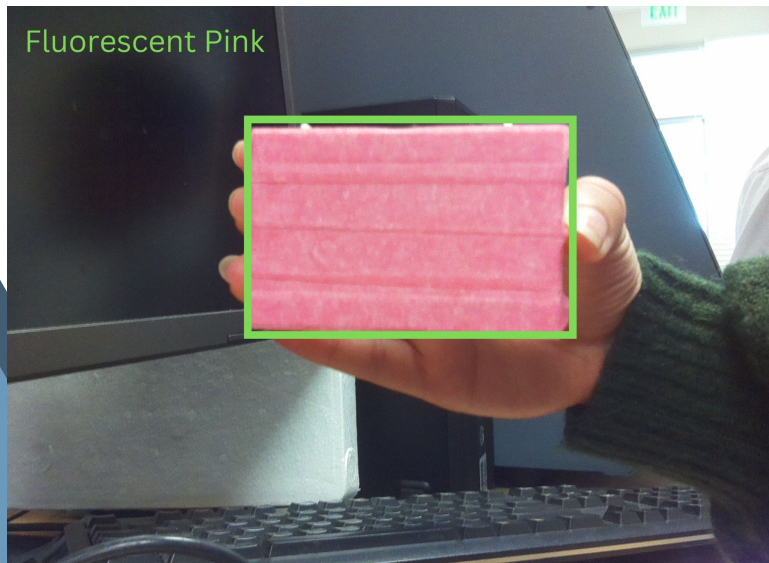


Figure 9.1: Vision Test

```
import cv2
import numpy as np

cap=cv2.VideoCapture(0)

while True:
    _, frame = cap.read()

    # Define the lower and upper bounds for fluorescent pink
    lower_range = np.array([139,0,139])
    upper_range = np.array([255,20,147])

    # Convert the frame to HSV color space
    hsv_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)

    hsv=cv2.cvtColor(frame,cv2.COLOR_BGR2HSV)

    mask = cv2.inRange(hsv_frame,lower_range,upper_range)
    _,mask1=cv2.threshold(mask,254,255,cv2.THRESH_BINARY)

    cnts,_=cv2.findContours(mask1,cv2.RETR_EXTERNAL,cv2.CHAIN_APPROX_NONE)

    for c in cnts:
        x=600
        if cv2.contourArea(c)>x:
            x,y,w,h=cv2.boundingRect(c)
            cv2.rectangle(frame,(x,y),(x+w,y+h),(0,255,0),2)
            cv2.putText(frame,"DETECT",(10,60),cv2.FONT_HERSHEY_SIMPLEX,0.6,(0,0,255),2)

    cv2.imshow("FRAME",frame)

    if cv2.waitKey(1)&0xFF==27:
        break

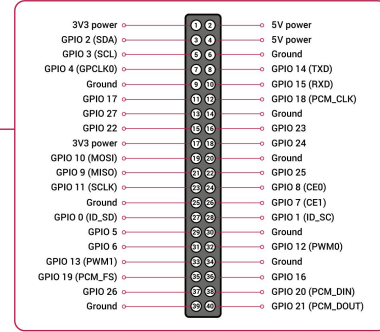
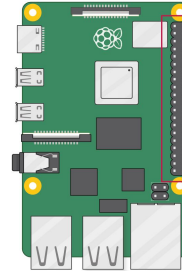
cap.release()
cv2.destroyAllWindows()
```

Figure 9.2: Vision Test Code

Programming Raspberry Pi with GPIO Pins

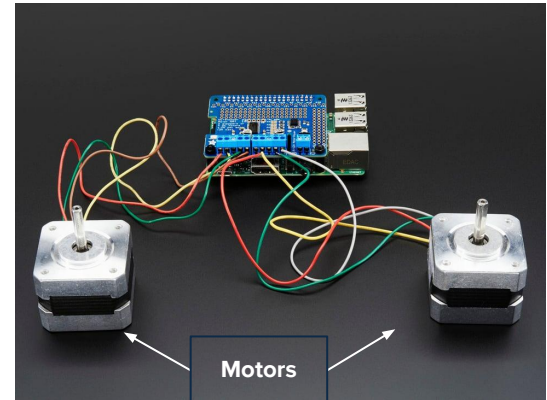
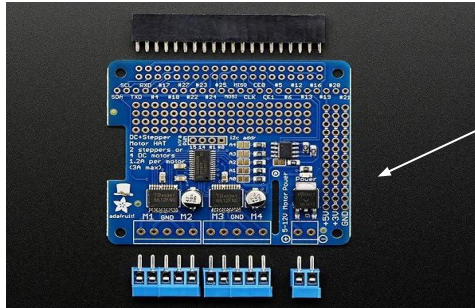
GPIO Pins

- **GPIO Pins (Raspberry Pi)** connect to **Adafruit Motor Hat**
- **Allows a transfer of electric signals** to reach the **motors**
- **Capable of controlling motors**



Raspberry Pi

Adafruit Motor Hat



Programming Motor System Connection

- Controls 4 Wheel motors and fan motors using PS4 Controller
- Primarily for testing, but a backup if in the worst case scenario autonomy fails.
- Condensed code used to control the motors



Figure 10.1: DUALSHOCK™4 Wireless Controller

```
import time
import board
from adafruit_motorkit import Motorkit
from pyPS4Controller.controller import Controller
kit = Motorkit(i2c=board.I2C())

def forward():
    kit.motor1.throttle = 1.0
    kit.motor2.throttle = 1.0
    kit.motor3.throttle = -1.0
    kit.motor4.throttle = -1.0
    print("forward")

class MyController(Controller):
    def on_up_arrow_press(self):
        forward()

controller = MyController(interface="/dev/input/js0", connecting_using_ds4drv=False)
controller.listen()
```



CLAW APPROACH

Claw Design Matrix

	Cost	Weight	Size	Efficiency	Precision	Practicality	Electrical Complexity	Manufacturing Complexity	Prgramming Complexity	Total
Roller	3.4	5.1	2.9	2.9	3.0	3.8	3.6	3.4	3.2	36.1
Pronged Gripper	3.9	3.7	3.6	3.7	3.5	3.3	3.5	3.4	3.5	38.2
Wire Intake	3.7	3.8	3.9	3.0	2.5	3.9	3.9	3.8	3.8	39.8
Trunk	2.7	2.5	2.6	3.4	3.1	2.2	2.3	2.7	2.2	27.5
Pneumatic Piston	2.9	3.1	3.0	3.2	3.1	2.9	2.1	3.4	3.1	33.9

Table 3: Design Matrix

Manufacturing Explanation

- Prototyping
 - PLA 3D Printing
 - Pulled prints from online libraries for claw types
- Problems:
 - Gear teeth did not interlock at all times
 - Very small surface area contact between claw and artifact
 - Print quality with CF-ABS was not the best → switch to PLA

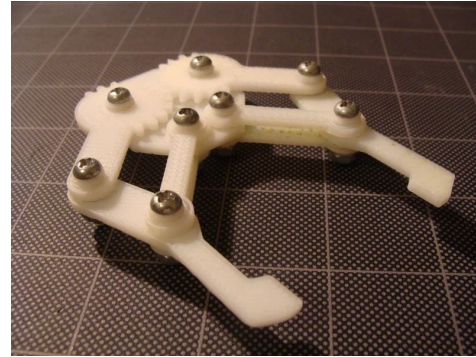
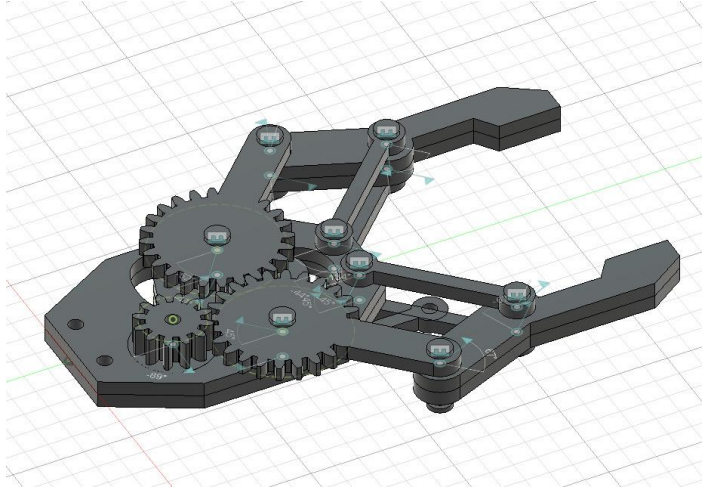


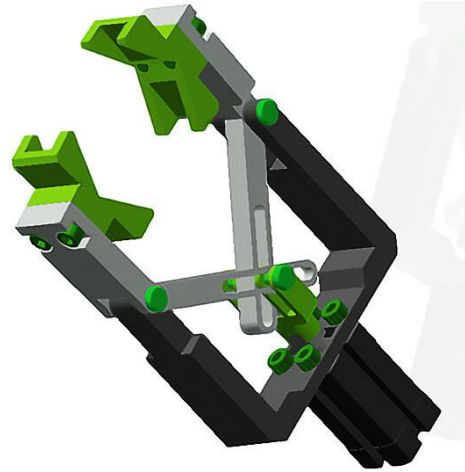
Figure 11: Claw Prototype



Potential Claw Options



Potential Claw Option
#1



Potential Claw Option
#2

F.2 Second Iteration

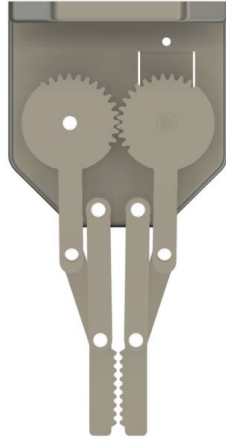


Figure 12.1: Top View

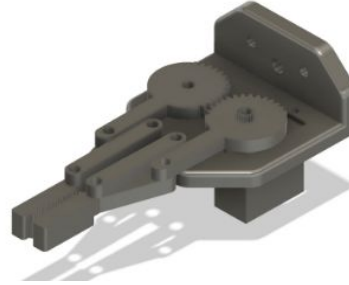


Figure 12.2: Isometric View

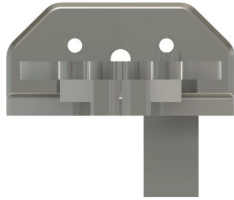


Figure 12.3: Front View

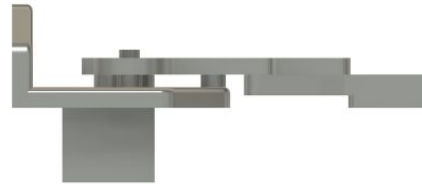


Figure 12.4: Side View

F.2 Iteration

- **Why the design was chosen?**
 - **Would work better with main module**
- **Improvements from first iteration**
 - **Gears now touch**
 - **More stable**
 - **Better printing quality**

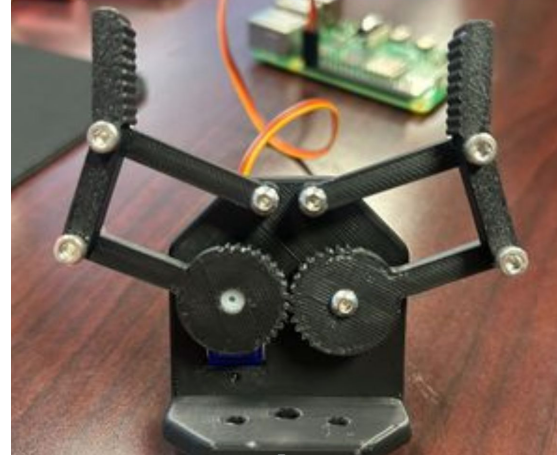


Figure 13 : Claw Prototype

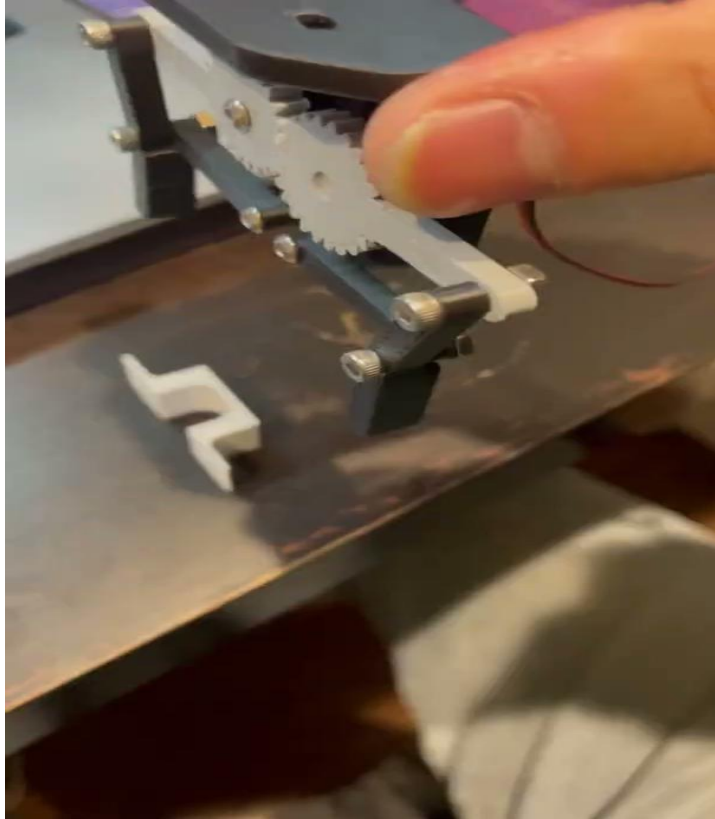
Programming Explanation

```
claw.py x
1 # Don't worry, I don't code in light mode
2 import gpiozero
3 import time
4
5
6 1 usage
7 class ClawModule:
8     def __init__(self, servo: gpiozero.Servo | int):
9         """
10         Claw
11         :param servo: The servo motor that controls the claw, or the GPIO pin that the servo is connected to
12         """
13         if isinstance(servo, int):
14             self.servo = gpiozero.Servo(servo)
15         else:
16             self.servo = servo
17
18 @
19 def __str__(self):
20     """
21     __str__(ClawModule) <==> str(ClawModule)"""
22     return "Claw Module"
23
24 1 usage
25 def open(self) -> None:
26     """Open the claw"""
27     self.servo.max()
28
29 1 usage
30 def close(self) -> None:
31     """Close the claw"""
32     self.servo.min()
33
34 1 usage
35 def detach(self) -> None:
36     """Detach the servo motor from the GPIO pins"""
37     self.servo.detach()
```

```
34 1 usage
35 def main():
36     # Set up the servo on GPIO 18
37     claw = ClawModule(18)
38     claw.open()
39     time.sleep(1)
40     claw.close()
41     time.sleep(1)
42     claw.detach()
43
44 try:
45     main()
46 except KeyboardInterrupt:
47     print("Program ended.")
48
49
50
```

Figure 14: Claw Code

Final Claw



Video 5: Claw Test



POLE CLIMBING MODULE PG

Pole Climbing Module - Drive System

- **8-Wheel Drive System**
 - Drives vertically up the pole
 - Belt & Pulley System
 - Utilizes 2 motors
 - Alternate 4-Motor Design
 - Pros
 - Powerful Design
 - Cons
 - Heavy Design
 - Balancing Issues
 - Cost

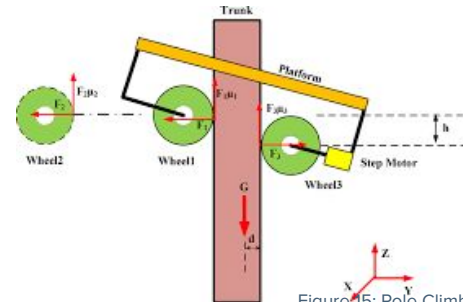
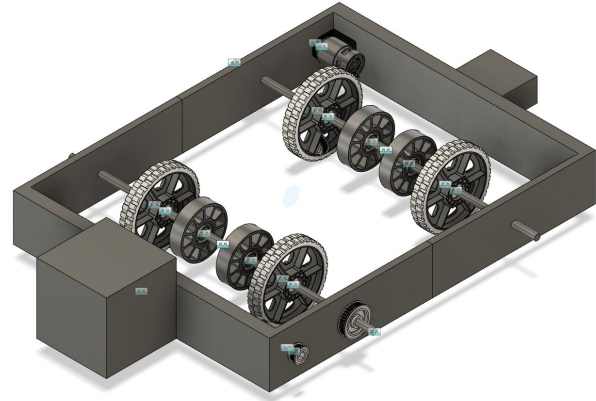
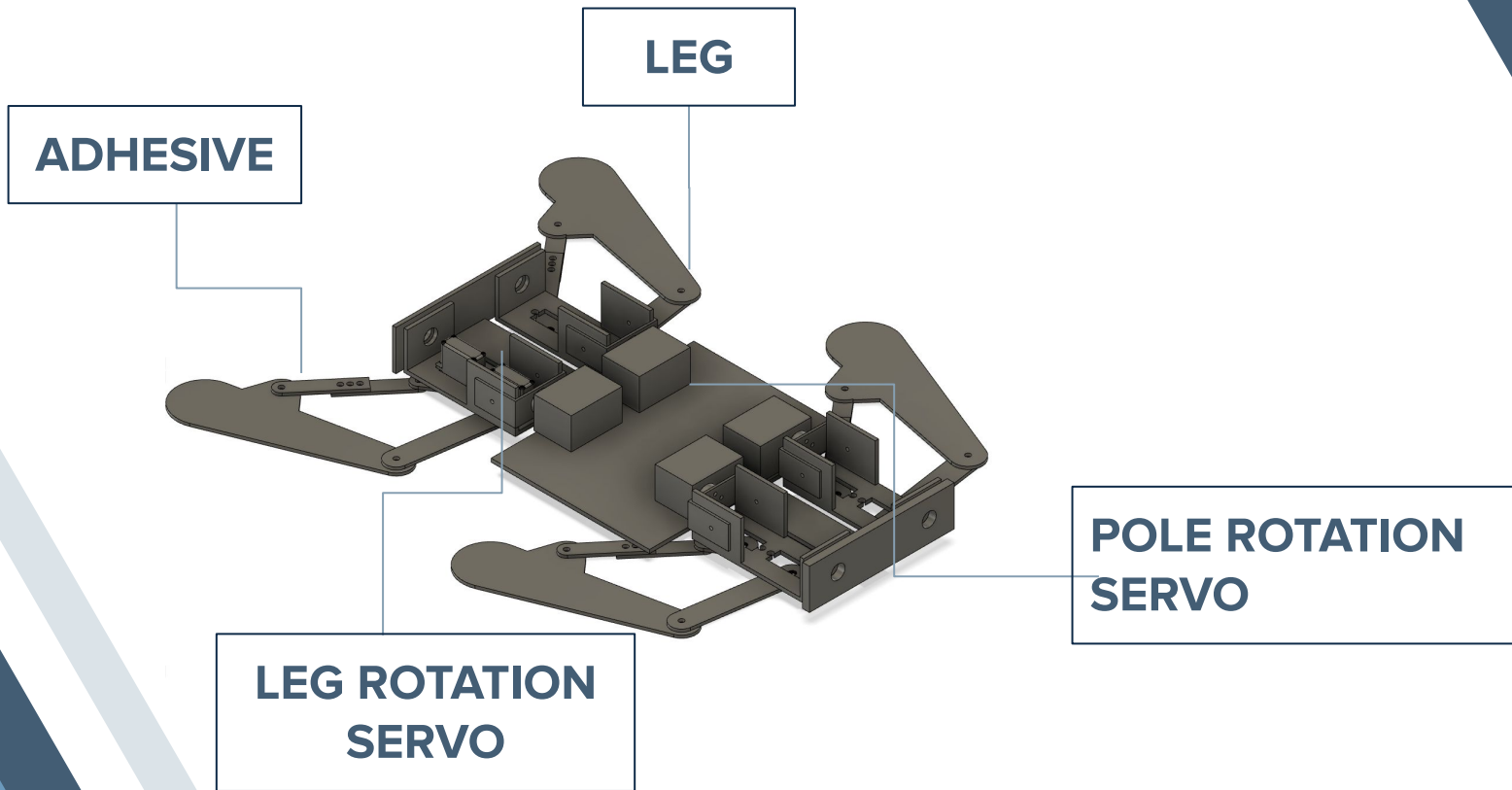


Figure 15: Pole Climbing Motion Diagram

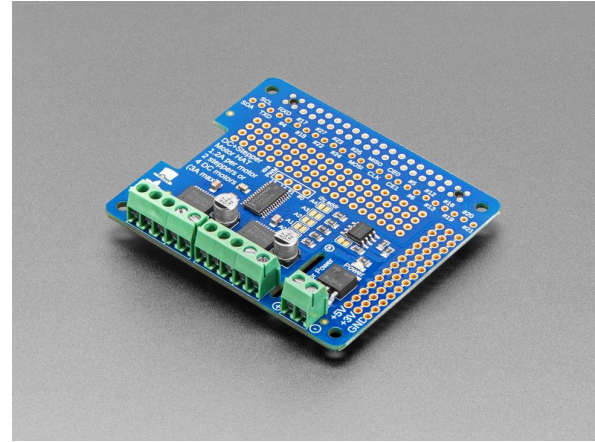
Pole Climbing Module - Panda Gecko

INSPIRED BY STANFORD



Sub-System Approaches

- Raspberry Pi Adafruit motor hat we will control 4 dc motors
- The Hat connects to the gpio pins on the raspberry pi allowing us to control the motors using
 - `adafruit_motorkit`



Mechanical Explanation - PG

- Legs/Five-Bar Linkage which allows:
 - Two degree of freedom mechanism
 - Connected in a closed chain
 - Utilizing two servo motors
- Servo Motors
 - Control two-degree freedom at joints A and B → controls angle of L2 and L5
 - L1 is grounded link, endpoint is D
 - On x-y plane, where angles 1 & 2 can be calculated as functions x,y

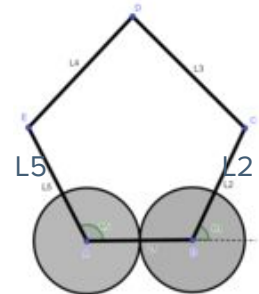
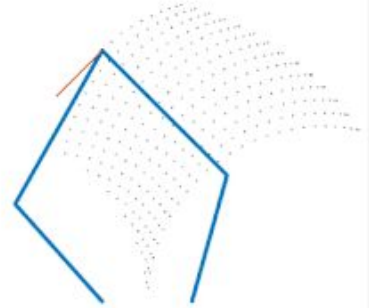


Figure 16: Five-Bar Linkage

Mechanical Explanation - PG

- Leg mechanism
- Rotations realized by the servo motors

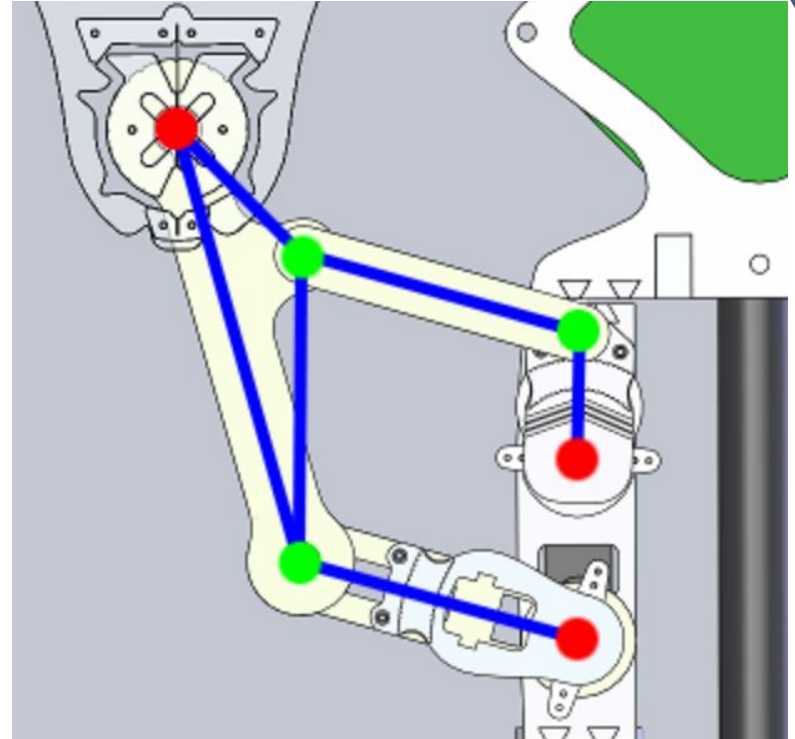


Figure 17: Principal Leg Mechanism

Mechanical Explanation - PG

- Degrees of freedom
- Servo utilization for elbow movement

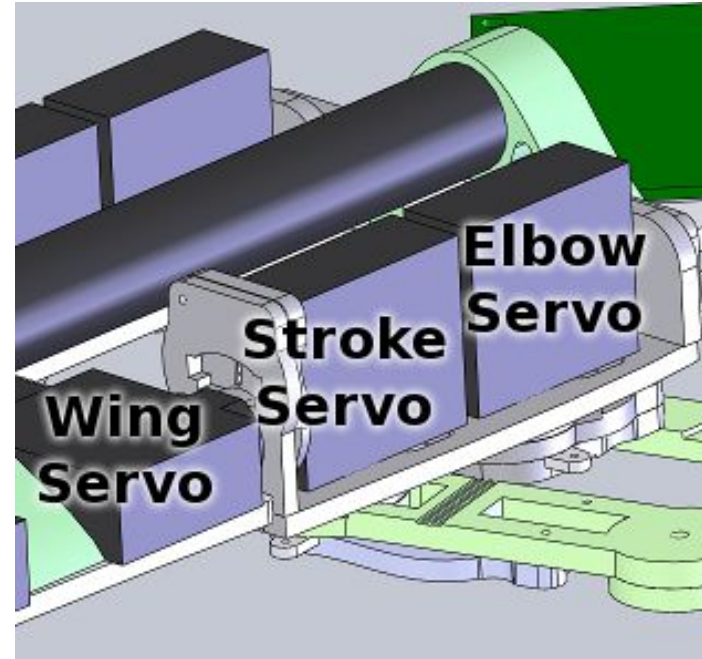


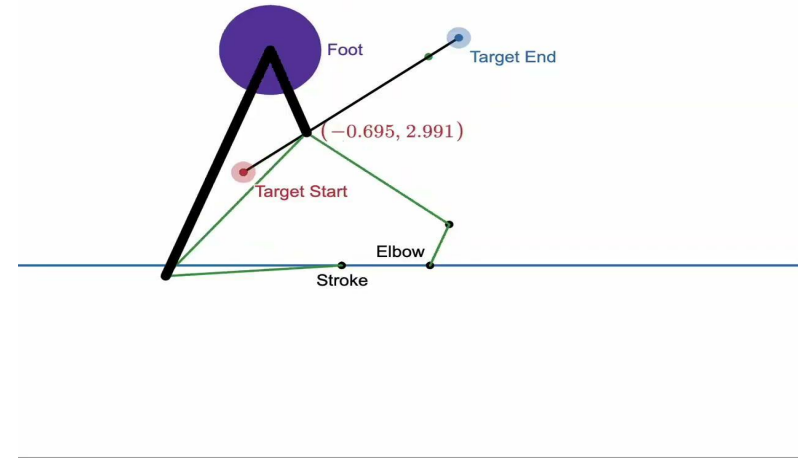
Figure 18: Wing's Degree of Freedom

Mechanical Visualization- PG



Programming Explanation - PG

- Inverse kinematics were calculated to allow legs to be controlled in the x-y plane
- Allows Panda Gecko to strafe



Manufacturing Explanation - PG

- **Main Body/Feet (In Progress/Design)**
 - **Acrylic**
 - One structure → less prone to cracking
 - More precise cuts
- **Motor/Servo Mounts**
 - **PLA 3D Printing**
 - Custom servo placement
 - “3D model”
 - Unable to achieve with laser cutting
 - Tensile Strength: 52.5 MPa
 - Flexural Strength: 65.9 MPa



Figure 19: 3D Printed Prototype of a Carriage

Manufacturing Explanation - PG

- **Vytaflex 60 or Tap Blue silicone**
 - **Molded in diffraction grating**
 - **Maximum adhesion in preferred sheared direction**
 - **No adhesion in the absence of a shear load**

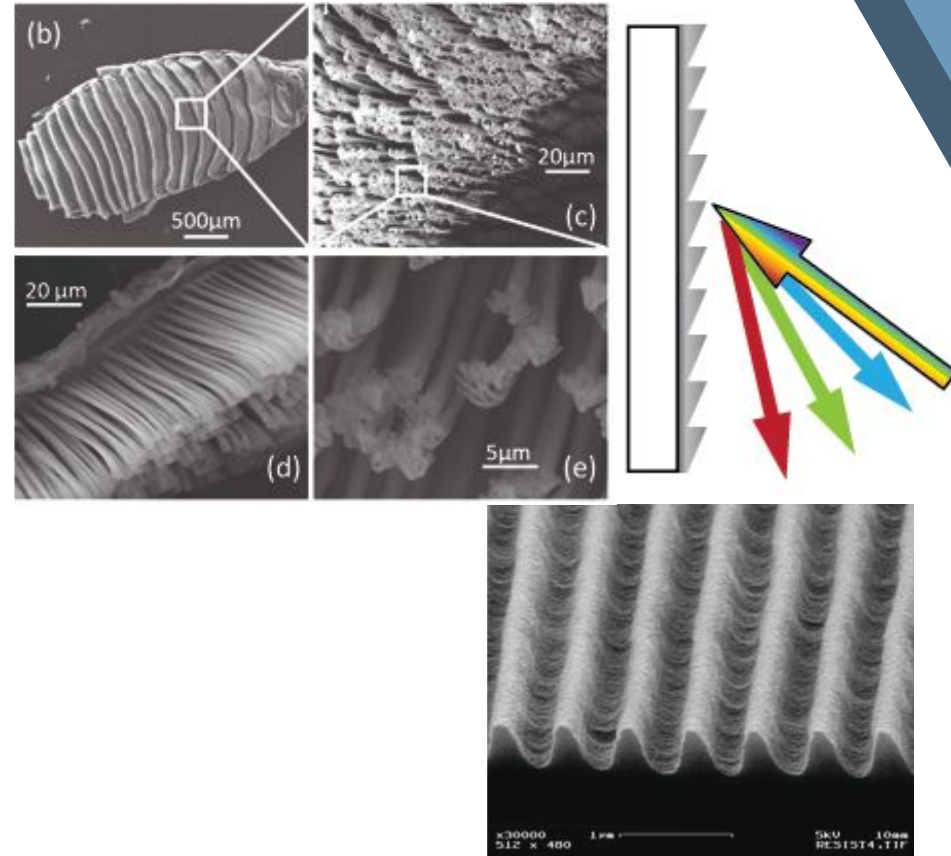


Figure 20: Microscopic close-up of adhesive

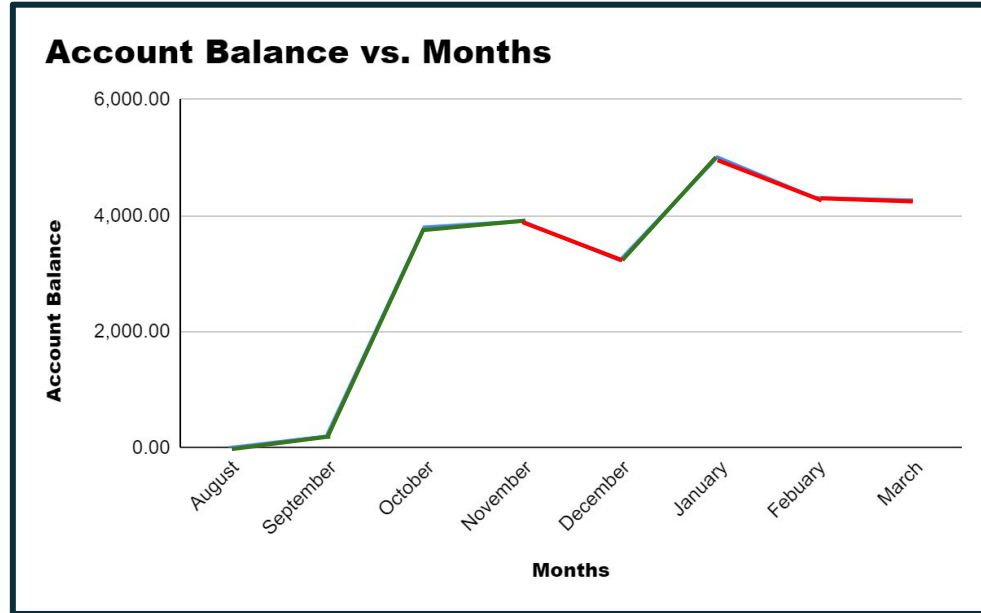
Mechanical Explanation - PG

- Tail
 - PLA 3D Printing
 - Reduces the pitch-back torque on the robot
 - Connected using strong magnets
- Detachable Mechanism
 - Magnetic connection allows the tail to break away from the body
 - Allows detach detach for storage and demonstrates tail's utility

The image features a white central area with the text "FINANCIAL REPORT" in a bold, dark blue, sans-serif font. This central area is framed by several overlapping geometric shapes in shades of blue and grey, creating a modern, abstract background.

FINANCIAL REPORT

Monthly Reports



Months	August	September	October	November	December	January	February	March	Overall
Gross Income	0.00	200.00	4,014.57	227.00	121.37	2,000.00	692.00	0.00	7,254.94
Deductions	0.00	0.00	(412.00)	(123.58)	(800.56)	(215.61)	(1,393.64)	(130.73)	(3,076.12)
Net Income	0.00	200.00	3,602.57	103.42	(679.19)	1,784.39	(701.64)	(130.73)	4,178.82
Account Balance	0.00	200.00	3,802.57	3,905.99	3,226.80	5,011.19	4,309.55	4,178.82	

Budget Allocation

Allocation Type	Allocated Budget	Actual Expense	Remaining Budget
Miscellaneous	\$1,600.00	-\$619.32	\$980.68
Robot	\$2,250.00	-\$1,549.46	\$622.40
Insurance	\$1,000.00	-\$651.86	\$348.14
Total	\$5,450.00	-\$2,820.64	\$1,951.22

Subteam	Allocated Budget	Actual Expense	Remaining Budget
Manufacturing	\$200.00	-\$53.96	\$146.04
Finance	\$700.00	-\$418.54	\$281.46
Media	\$300.00	-\$123.58	\$176.42
Electrical	\$1,000.00	-\$574.66	\$425.34
Programming	\$300.00	-\$300.00	\$0.00
Mechanical	\$750.00	-\$698.98	\$51.02
Interns	\$200.00	\$0.00	\$200.00
Publications	\$400.00	-\$77.20	\$322.80
Systems	\$0.00	\$0.00	\$0.00
Insurance	\$1,000.00	-\$651.86	\$348.14

Bill of Materials (Mechanical)

Mechanical							
Name	Description	Vendor	Unit Price	Base Unit	Quantity to Order	Total Units	Total Price
Battery - 12v	Batteries	Amazon	\$5.21	2	1	2	\$5.21
12v-5v converter	Raspberry Pi	Amazon	\$8.99	1	1	1	\$8.99
Metric Screws	Assembly	Amazon	\$24.99	1	1	1	\$24.99
Battery Holder	Organization	Amazon	\$7.99	10	1	10	\$7.99
Magnets	Mechanical	Amazon	\$6.99	9	1	9	\$6.99
Electromagnets	Mechanical	Buy Magnets	\$27.55	1	2	1	\$55.10
3602 Series Press-Fit Wheel (3mm D-Bore, 80mm Diameter) - 2 Pack	Wheels	Servocity	\$9.49	2	2	4	\$27.97
DC Toy / Hobby Motor - 130 Size	DC Motors	Adafruit	\$1.95	1	8	8	\$15.60
DC Motor Plastic Mount - 130 Size / 20mm Diameter	DC Motors	Adafruit	\$0.75	1	8	8	\$6.00

Bill of Materials (Electrical)

Electrical							
Name	Description	Vendor	Unit Price	Base Unit	Quantity to Order	Total Units	Total Price
HS-S2MG Standard Metal Gear Micro Servo	Responsible for the movement of the elbow, stroke, and wing. All move the arm up and down.	Amain Hobbies	\$23.99	1	13	13	\$311.87
6S Motor Speed Controller	Speed Controller	Amazon	\$22.92	1	1	1	\$22.92
XT90 Connectors	Connectors	Amazon	\$11.99	1	1	1	\$11.99
LiPo 2000 6S 22.2v Battery Pack	Batteries	Maxamps	\$119.99	1	1	1	\$128.24
Wheel gearboxes + motor-12V (CLIMB)	Climb Module	Amazon	\$11.99	1	16	16	\$211.52
Stepper Motor	Motors	Amazon	\$26.35	1	1	1	\$26.35
Connector Plugs	Connectors	Amazon	\$7.99	12	1	12	\$7.99
Yaohappy 5Pair Black Amass XT90 XT-90 Male Female Bullet Connectors Power Plug with Heat Shrink for RC Lipo Battery Motor	Connectors	Amazon	\$12.99	10	1	10	\$12.99
heat shrink	Soldering	Amazon	\$6.99	1	1	1	\$6.99
Flashhobby A2807-1500KV	Brushless Motors	Bang Good	\$16.99	1	1	1	\$16.99
Flashhobby A2807-1800KV	Brushless Motors	Bang Good	\$16.99	1	1	1	\$16.99
brushless esc (CLIMB)	Climb Module	Xoar	\$62.99	1	1	1	\$62.99
brushless motors + propeller (CLIMB)	Climb Module	Xoar	\$102.88	1	1	1	\$102.88
Battery	Battery	Amazon	\$19.79	1	1	1	\$19.79
Wheel gearboxes + motor-12V (CLIMB)	Climb Module	Amazon	\$11.99	1	4	4	\$47.96
Propellers	Smaller Propellers	Amazon	\$12.99	3	1	3	\$12.99
FrSky L9R Long Range Receiver for FrSky Taranis X9DPLUS	Long Range Receiver	Amazon	\$29.99	1	1	1	\$29.99

Bill of Materials (Manufacturing)

Manufacturing							
Name	Description	Vendor	Unit Price	Base Unit	Quantity to Order	Total Units	Total Price
Diffraction Grating sheet	For adhesive	amazon	\$19.95	1	1	1	\$19.95
Sommoth-On VytaFlex 60 - Liquid Urethane Rubber	For adhesive	ebay	\$36.02	1	2	2	\$72.04
TAP Silicone RTV Mod-Makin System	TAP Plastics	Tap Plastics	\$94.75	1	1	1	\$94.75
Screws	Metric Screws	Amazon	\$9.99	1	1	1	\$19.98
Acrylic Sheet	12 by 16 6mm	Amazon	\$9.99	2	1	2	\$19.98
Acrylic Sheet	12 by 16 Sheets 3mm	Amazon	\$18.98	2	1	2	\$20.93
Glue	Glass Glue for Adhesive	Amazon	\$9.99	1	1	1	\$9.99

Bill of Materials (Programming)

Programming							
Name	Description	Vendor	Unit Price	Base Unit	Quantity to Order	Total Units	Total Price
Traxxas 3496 Waterproof Electronic Speed Control - VXL8S ESC Xmaxx XRT	Speedcontroller	Ebay	\$117.26	1	1	1	\$117.26
Raspberry Pi 5 8GB Starter Kit	Raspberry Pi	Canakit	\$154.95	1	1	1	\$154.95
Dupoint Wire	Multicolored Dupoint Wires	Amazon	\$6.98	120	1	120	\$6.98
Duct Tape	3M	Amazon	\$7.02	1	1	1	\$7.02
Raspberry Pi Cable	Raspberry Pi	SparkFun	\$1.00	1	1	1	\$1.00
Adafruit DC & Stepper Motor Bonnet for Raspberry Pi	Stepper Motor	Adafruit	\$22.50	1	2	2	\$45.00
GPIO Header for Raspberry Pi A+/B+/Pi 2/Pi 3/Pi 4/Zero - 2x20 Female Header	Raspberry Pi	Adafruit	\$1.50	1	2	2	\$3.00
3602 Series Press-Fit Wheel (3mm D-Bore, 32mm Diameter) - 2 Pack	Wheels	Servocity	\$4.99	2	2	4	\$9.98

Bill of Materials (Miscellaneous)

Finance

Name	Description	Vendor	Unit Price	Base Unit	Quantity to Order	Total Units	Total Price
Trays	2 LB Trays	Amazon	\$11.99	100	1	100	\$11.99

Media

Name	Description	Vendor	Unit Price	Base Unit	Quantity to Order	Total Units	Total Price
Banner	Presentation	Vistaprint	\$1.00	1	1	1	\$87.58
Business Cards	Presentation	Canva	36	200	1	200	\$36.00

Publications

Name	Description	Vendor	Unit Price	Base Unit	Quantity to Order	Total Units	Total Price
Presentation Ringed Book	SPS Document	Office Depot	\$38.60	1	2	2	\$77.20

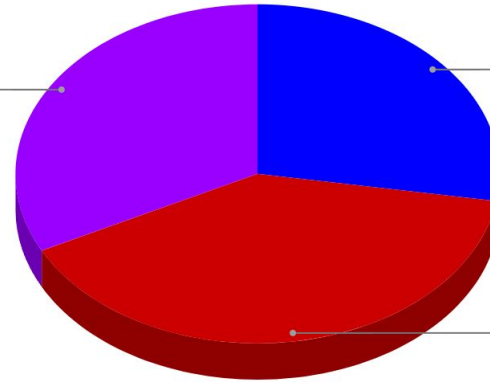
Funds Breakdown

\$7000
Gross
Income

March
February
January
December

Income

Donors
32.5%



Grants
27.6%

Fundraisers
40.0%

Fundraisers, Sponsors, Grants

Past Fundraisers:

- Christmas Candy Grams
- Pasta
- Croissants and Hot Cocoa

Future Fundraisers:

- Nachos
- Montera Apparel
- Spam musubi
- 7 Leaves Boba

Sponsors:

- Tacos El Charro
- AT&T

Grants:

- Long Beach Unified School District
- Kinecta Teacher Grant



AT&T

Work Cited

MANUFACTURING:

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9572023/> (PLA Research)

<https://www.matweb.com/search/datasheet.aspx?bassnum=O1303&ckck=1> (ACRYLIC Research)

MANUFACTURING DOCUMENTATION:

<https://lbschools.sharepoint.com/:p:/s/EDDPeriod3/EdkPP3E4yX1HnEtMnx1B3JcBxDsQs3ej99bbCSs0-y0Nag?e=fcHtez>

PANDA GECKO: INSPIRED BY STANFORD'S RiSE PROJECT

<http://bdml.stanford.edu/twiki/bin/view/Rise/WebHome>

CLIMB ENGR 7 RESEARCH

https://59fc7c2c-42b4-4795-85ae-6204398ef245.filesusr.com/ugd/dbe9e7_bea249ff8f614021a2bb76f8630b201a.pdf



THANK YOU